

ECE435: Medical Imaging Project 3

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Introduction

This project explores the application of convex optimization and compressed sensing to anatomical MRI data. Taken from the OpenNeuro dataset, T1 and T2-weighted NIfTI are utilized to evaluate image compression performance across Fourier and Wavelet domains (Haar, Daubechies 4, and Coiflet 3) to exploit the idea of sparsity. Finally, a compressed sensing experiment is performed using a gradient descent algorithm that reconstructs images from randomly undersampled measurements, exploring the balance between data fidelity and sparsity-based regularization.

Part I: Guiding Computations

Figure 1 shows cross sections of a patient CSI1's MRI images taken along the sagittal, coronal, and axial planes for both the T1 and T2 weighted data.

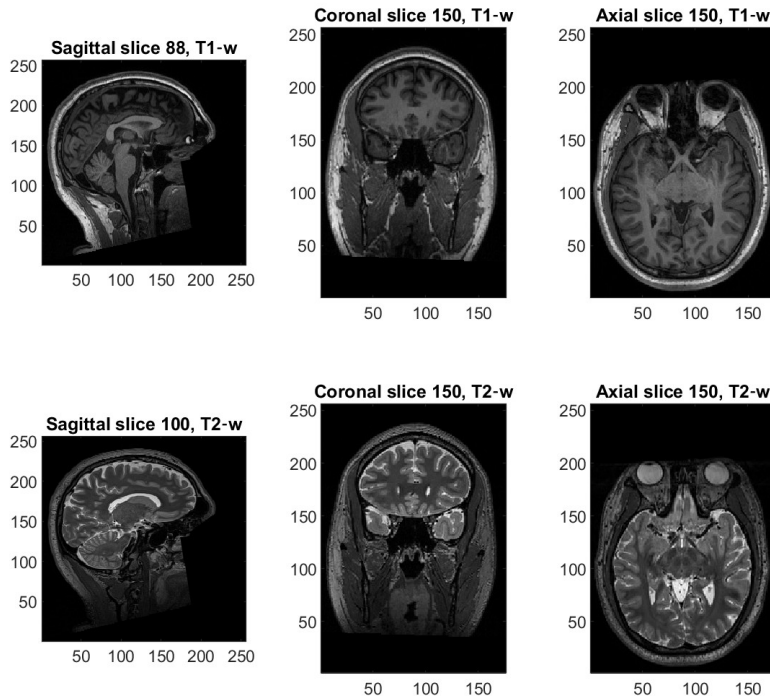


Figure 1: Cross-sections taken at $x = 88$, $y = 150$, and $z = 150$. The images are $176 \times 256 \times 256$ with pixel sizes $dx = dy = dz = 2\text{mm}$, hence the physical dimensions are $35.2\text{cm} \times 51.2\text{cm} \times 51.2\text{cm}$.

From the table of T1 and T2 values provided, it appears that tissues with shorter relaxation times appear brighter in the T1-weighted image. This is evidenced by the white matter appearing the brightest, most apparent in the coronal and axial slices. Conversely in the T2-weighted images, tissues with a longer relaxation time appear brighter. This is because the grey matter and cerebrospinal fluid appear brighter than white matter, almost as if the colors were inverted.

Looking at the JSON files for T1, the given values are: echo time (TE) = 0.00197, inversion time (TI) = 0.9, and repetition time (TR) = 2.3. For T2, the given values are: echo time (TE) = 0.422 and repetition time (TR) = 3.

TE is the delay between the excitation pulse and when the MR signal is measured. This controls T2 weighting because transverse magnetization decays so a long TE allows tissues with short T2 values to lose signal while tissues with long T2 values remain bright which is evidenced by the brightness of the spinal fluid.

TR is the time between repeated excitations. If TR is short, tissues do not all have enough time to fully recover before the next pulse and so tissues with shorter T1 relaxation times produce stronger signal, while tissues with long T1 appear darker. This is evidenced by the spinal fluid being very dark in the T1-weighted images.

TI is used only when there is an inversion pulse, which is why it is not provided in the T2 JSON file. The inversion pulse first flips the longitudinal magnetization, and then the scanner waits before imaging. During this waiting period, tissues recover at different rates depending on their T1 values. Tissues that recover more longitudinal magnetization by the time of measurement produce a stronger signal and appear brighter, while tissues that have recovered less appear darker. In the T1 images, the spinal fluid is dark which matches with the longer T1 time.

Based on the values provided in the table, the unitless times provided in the JSON files are mostly likely in seconds. This would align with common MRI parameters and also with the fact that the tissue relaxation values in the table are given in milliseconds.

Part II: Axillary Functions

These functions are provided in the MATLAB file.

Part III: Fourier Domain Compression

The chosen guided image was the T1-weighted sagittal cross section ($x = 88$). Figure 2 shows the magnitude spectrum of the guiding image. Figure 3 shows the histogram of the coefficient magnitudes in the guiding image's Fourier magnitude spectrum. The majority of the coefficients are small in magnitude, hence the image is sparse in the Fourier domain.

Figure 4 shows the results of compressing and reconstructing by removing the lowest s% magnitude coefficients. As expected, as more coefficients are removed (compressed), the reconstructed images become noticeably blurrier and the MSE compared to the original as ground truth also increases. The fact that the majority of coefficients can be removed and still maintain a degree of clarity highlights the sparsity of the image in the Fourier domain. However, the MSE value is quite high indicating this may not be the best method of compression.

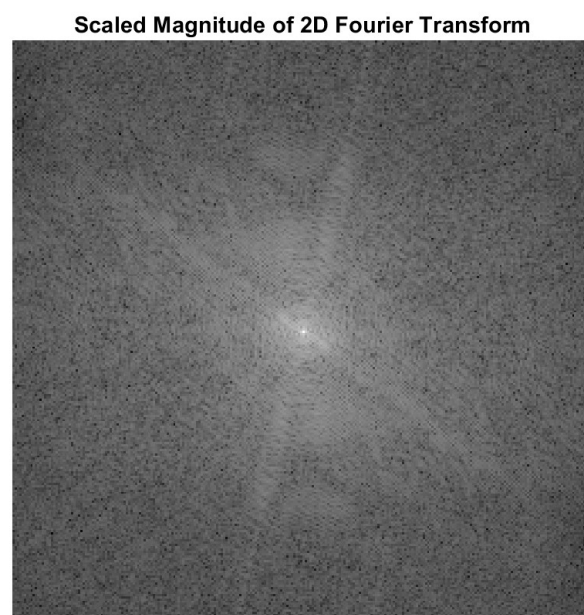


Figure 2: Magnitude spectrum of the guiding image.

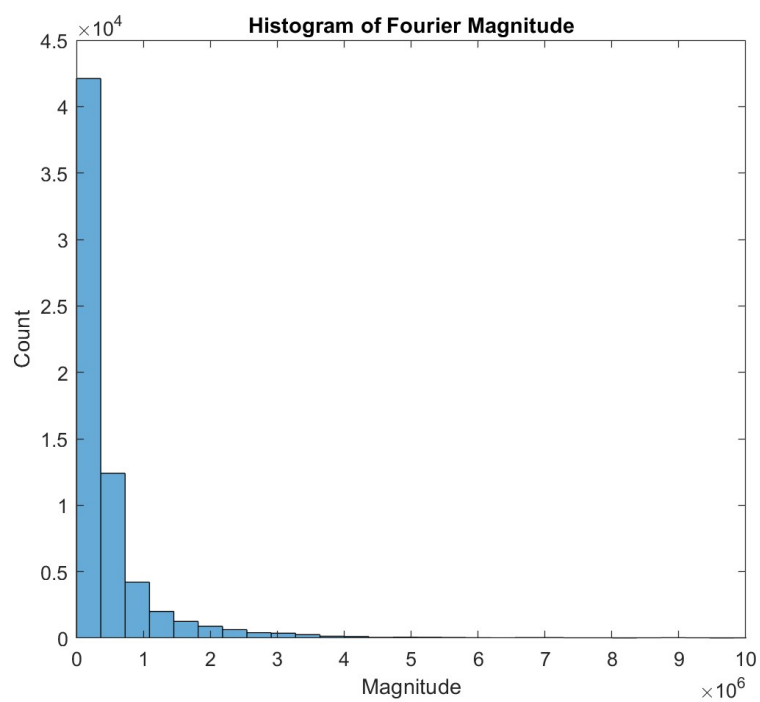


Figure 3: Histogram of magnitudes in guiding image magnitude spectrum.

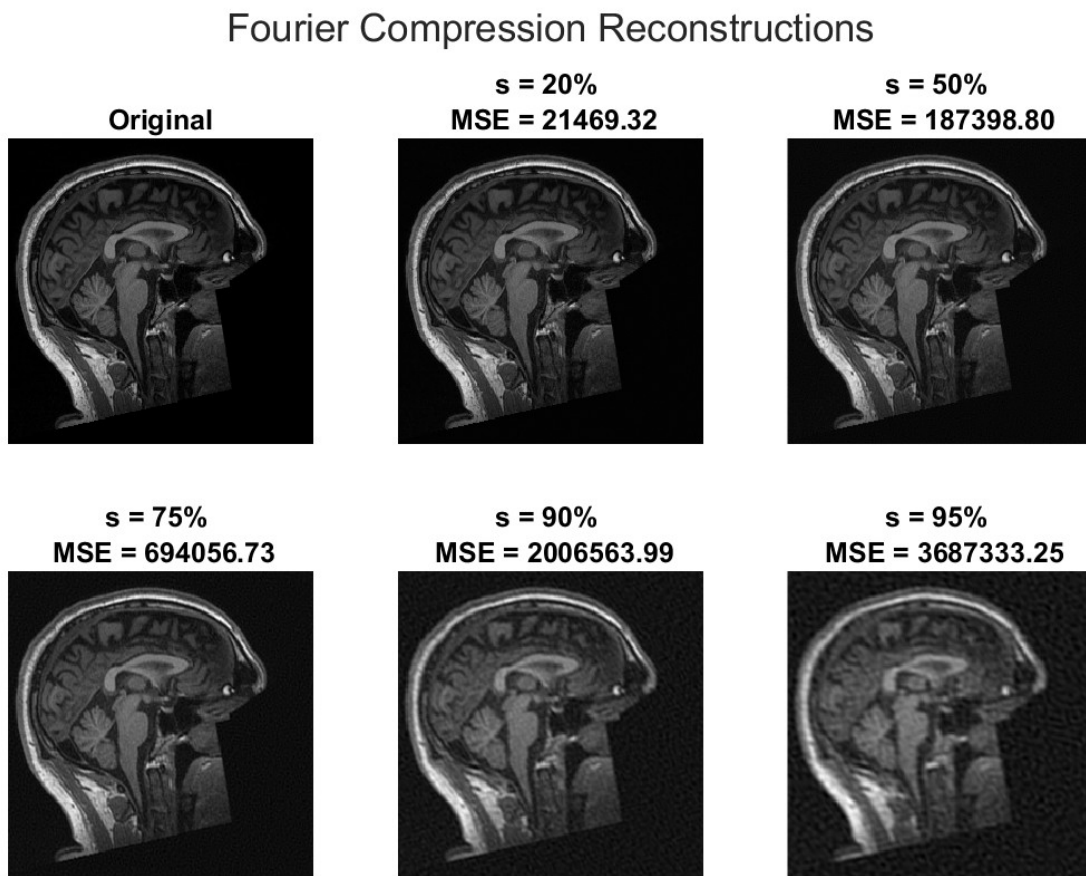


Figure 4: Fourier reconstructed images with various percentage of coefficients removed.

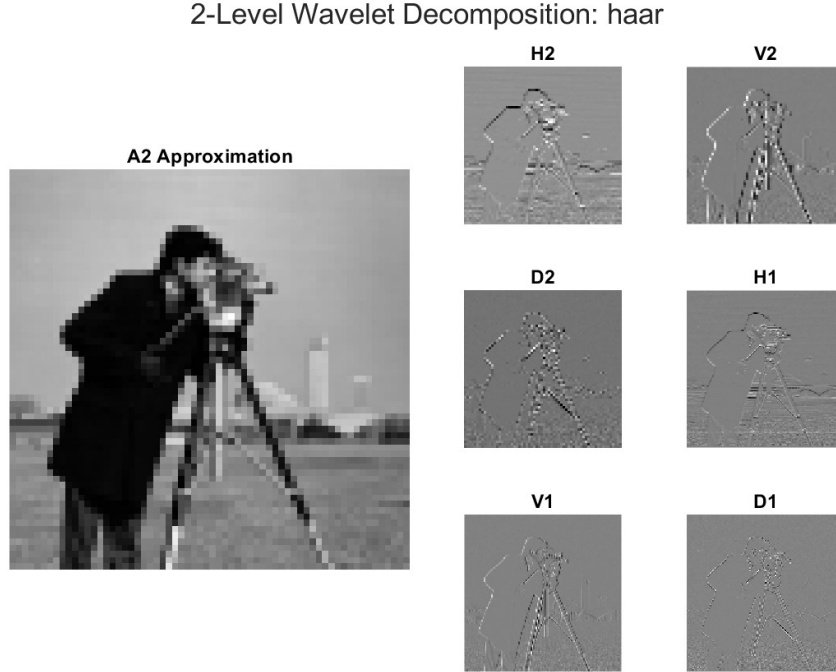


Figure 5: 2-level wavelet decomposition using the Haar basis.

Part IV: Wavelet Domains

Using the function written for Part 2 of this report to represent `cameraman` image in the Haar, Daubechies 4 (`db4`), and Coiflet 3 (`coif3`) wavelet basis, the 2-level wavelet decomposition is shown in the Figures 5, 6, 7.

The standard basis element (SBE) image is completely localized in space, since it is nonzero at only one pixel. Its Fourier magnitude, however, is broadly distributed across the frequency domain, showing that a spatial impulse contains all frequencies. The 2D wavelet basis elements are also spatially localized, but unlike the impulse, they occupy small neighborhoods rather than a single pixel. The Haar wavelet appears blocky and most compact, while the Daubechies-4 and Coiflet-3 wavelets are smoother and more oscillatory. Their Fourier magnitudes are spread across frequency bands rather than uniformly across all frequencies. Thus, wavelet basis elements provide joint space-frequency localization: they are localized in space while also concentrating their frequency content more than a single-pixel impulse.

The Haar basis elements are the most spatially localized, appearing as very compact, block-like responses around the impulse. In Figure 8 and 9, the Fourier magnitudes appear almost uniform gray across, meaning poor frequency localization. This is expected because the Haar wavelets are simply rectangular pulses. The `db4` wavelet coefficient images are still localized near the delta impulse, but they are slightly less point-like than Haar. In the Fourier magnitude plots, the `db4` basis shows visible bright and dark regions, reflecting that `db4` captures frequency information more clearly, with different patterns appearing in the horizontal, vertical, and diagonal components. Similarly, the Fourier magnitude plots of the `coif3` basis show clearer large-scale patterns, with bright and dark regions forming bands. This means the `coif3` basis represents the frequency content in a more organized way.

2-Level Wavelet Decomposition: db4

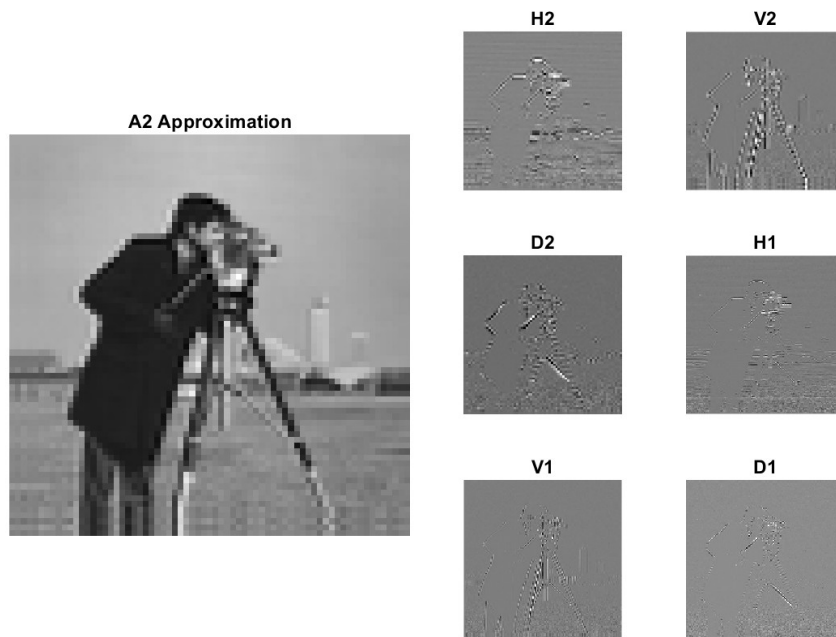


Figure 6: 2-level wavelet decomposition using the db4 basis.

2-Level Wavelet Decomposition: coif3

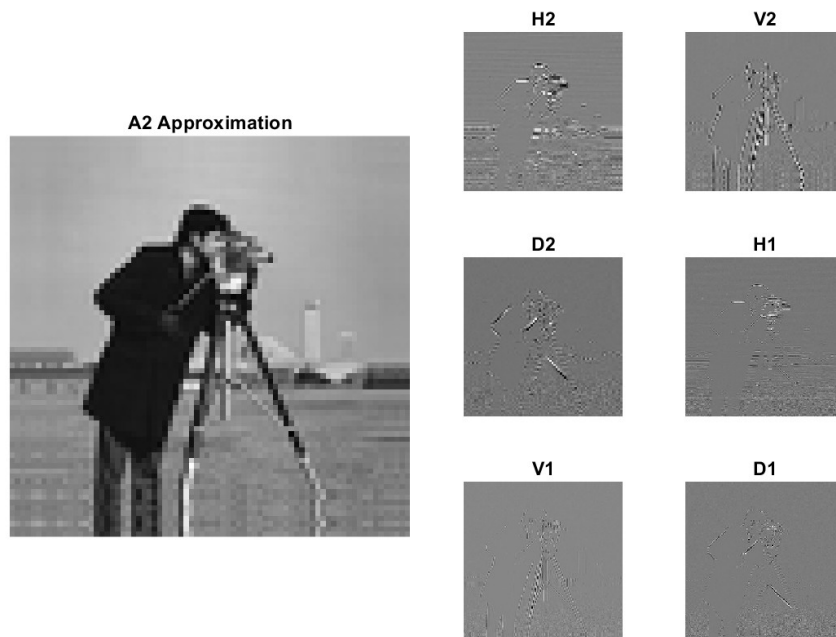


Figure 7: 2-level wavelet decomposition using the coif3 basis.

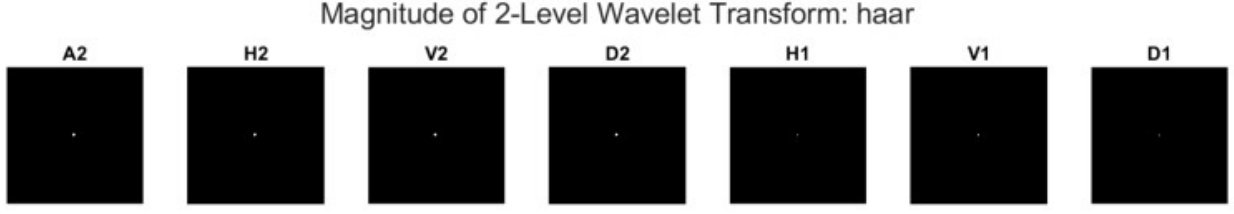


Figure 8: Magnitudes of the 2-level wavelet decomposition coefficients of SBE using the Haar basis.

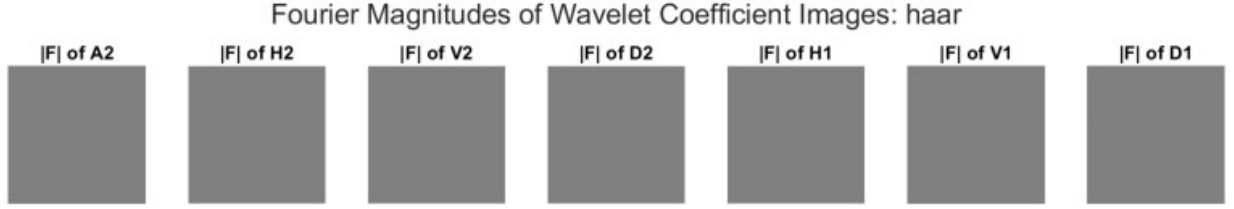


Figure 9: Fourier Magnitudes of the 2-level wavelet decomposition coefficients of SBE using the Haar basis.

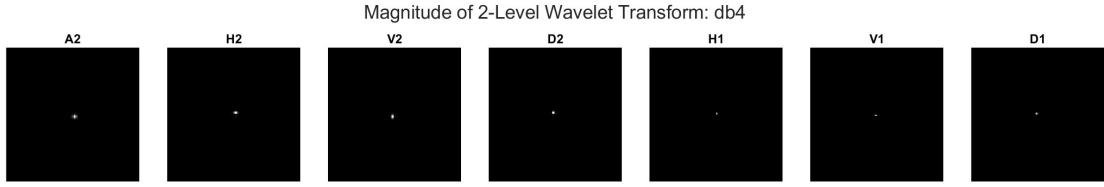


Figure 10: Magnitudes of the 2-level wavelet decomposition coefficients of SBE using the db4 basis.

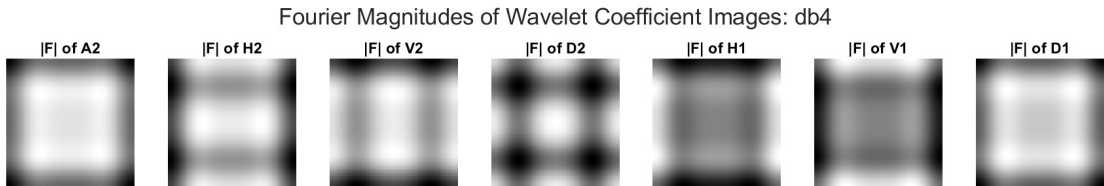


Figure 11: Fourier Magnitudes of the 2-level wavelet decomposition coefficients of SBE using the db4 basis.

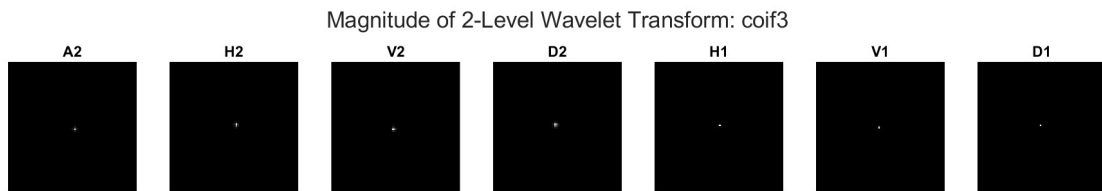


Figure 12: Magnitudes of the 2-level wavelet decomposition coefficients of SBE using the coif3 basis.

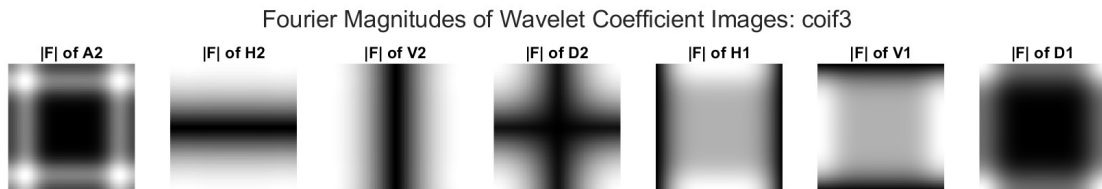


Figure 13: Fourier Magnitudes of the 2-level wavelet decomposition coefficients of SBE using the coif3 basis.

2-Level Wavelet Decomposition: haar

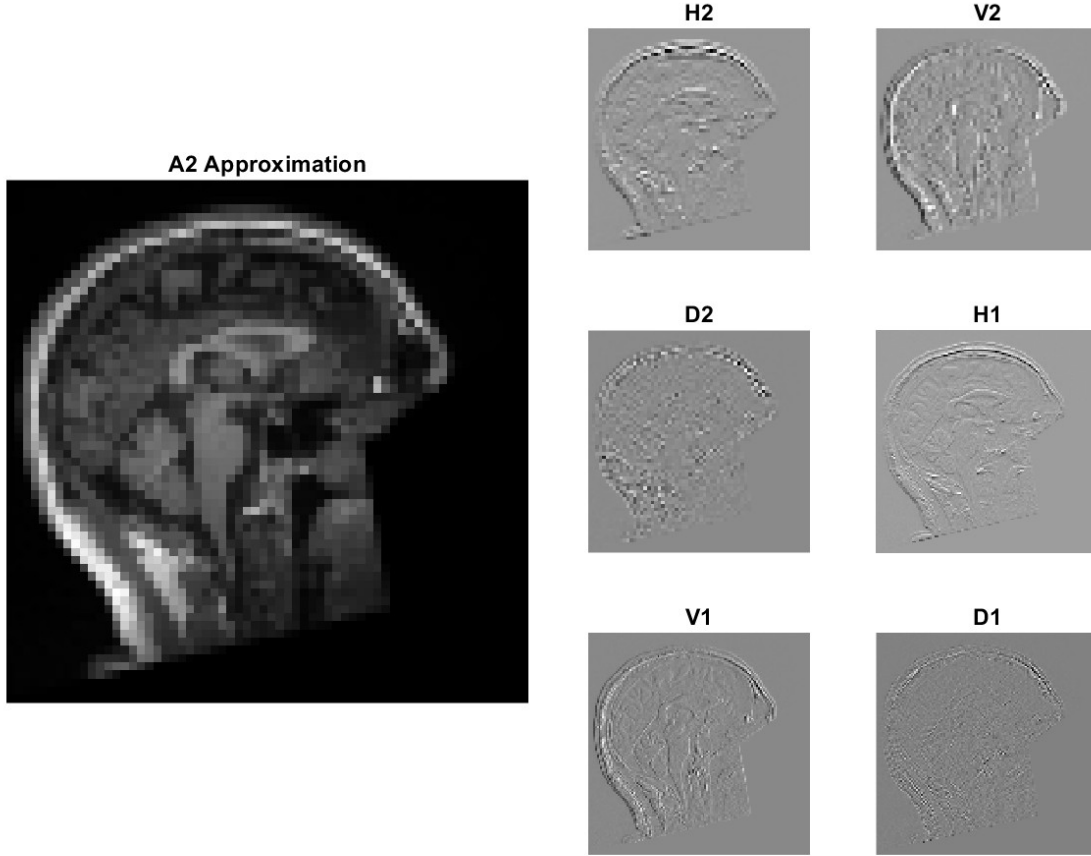


Figure 14: 2-Level Wavelet Decomposition of Guiding Image with Haar Basis.

Next, the 2-level decompositions of the guiding image in these three bases are shown in Figure 14, 15, 16.

The histograms for the total set of coefficients in these three bases after 1-level, 2-level and 3-level decomposition are shown in Figures 17, 18, 19.

In all 9 histograms, the distributions are highly concentrated near zero indicating the image is sparse in each of the three wavelet domains. Among the three tested wavelet bases and across the three decomposition levels, `coif3` appears to give the sparsest representation of the guiding image. Its coefficient histograms show the highest concentration near zero, with relatively fewer medium and large magnitude coefficients compared with `Haar` and `db4` across all 3 levels. As such, `coif3` appears to be the best candidate among the tested bases for compression or compressed sensing.

Next, the lowest 10% of the coefficients are zeroed for the guiding image. Using the `coif3` wavelet basis to reconstruct the image, the results are shown in Figure 20. The reconstruction is visually nearly identical to the original image, with negligible reconstruction error. The MSE is also incredibly small (basically 0).

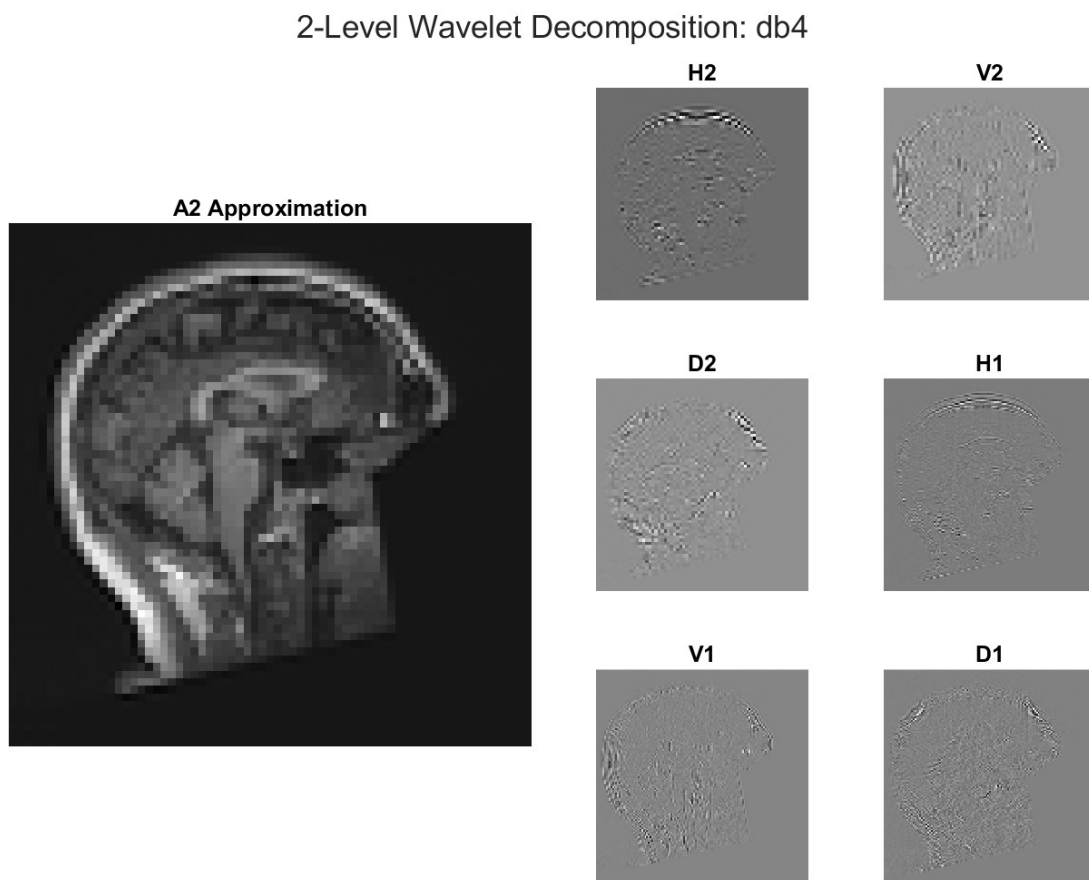


Figure 15: 2-Level Wavelet Decomposition of Guiding Image with db4 Basis.

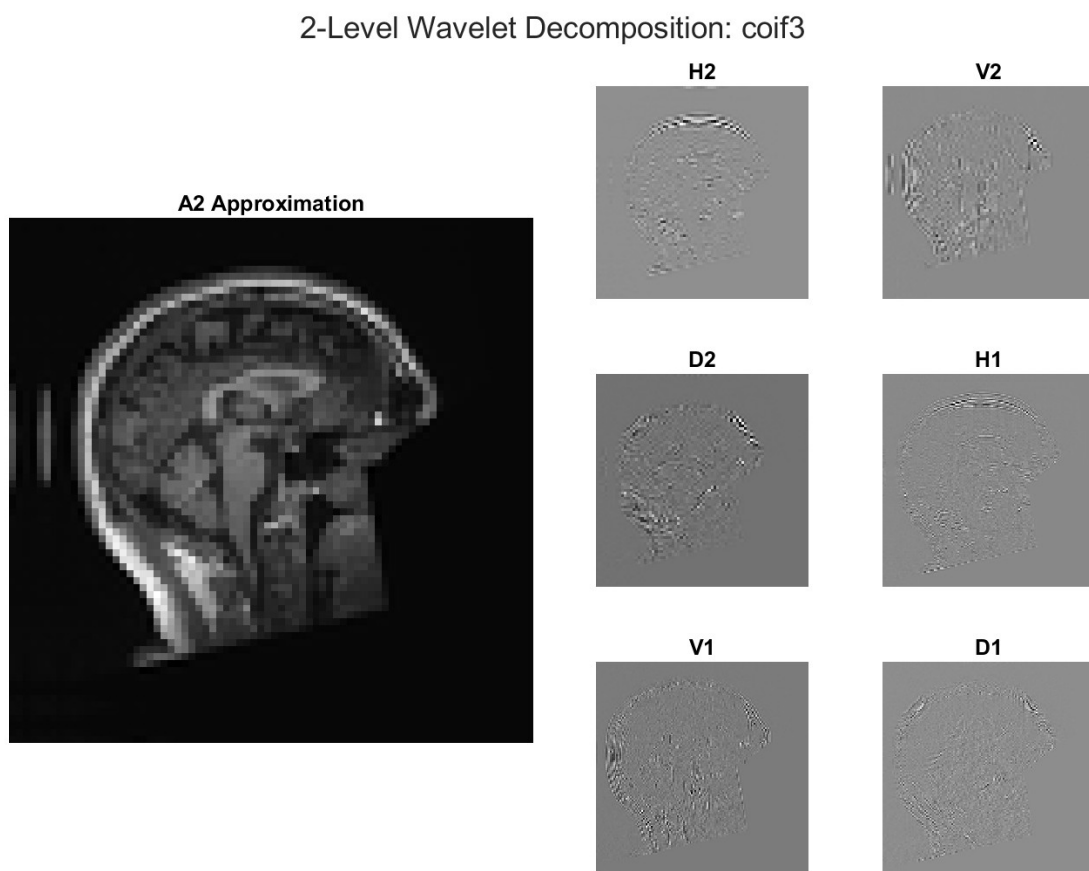


Figure 16: 2-Level Wavelet Decomposition of Guiding Image with coif3 Basis

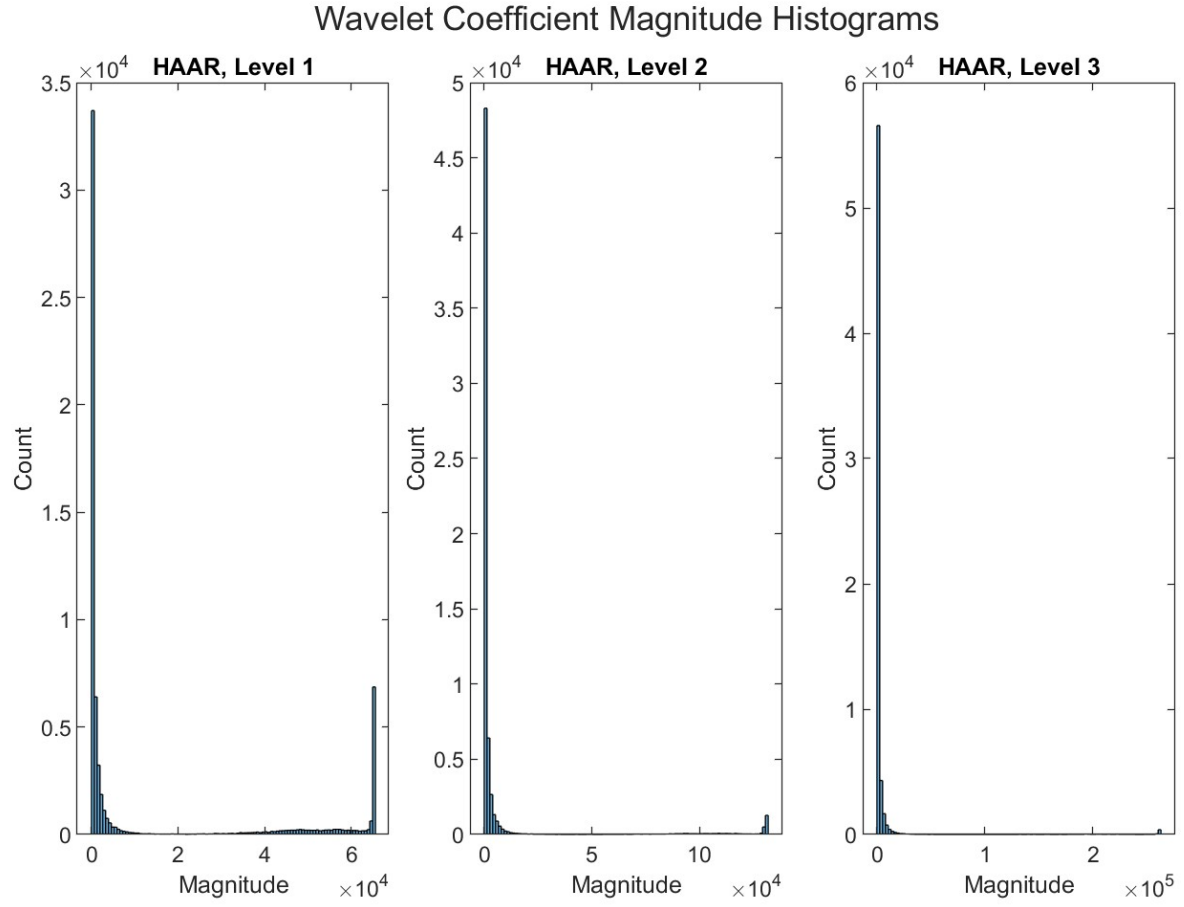


Figure 17: Wavelet coefficient magnitude histograms for the guiding image using the Haar wavelet basis at decomposition levels 1, 2, and 3.

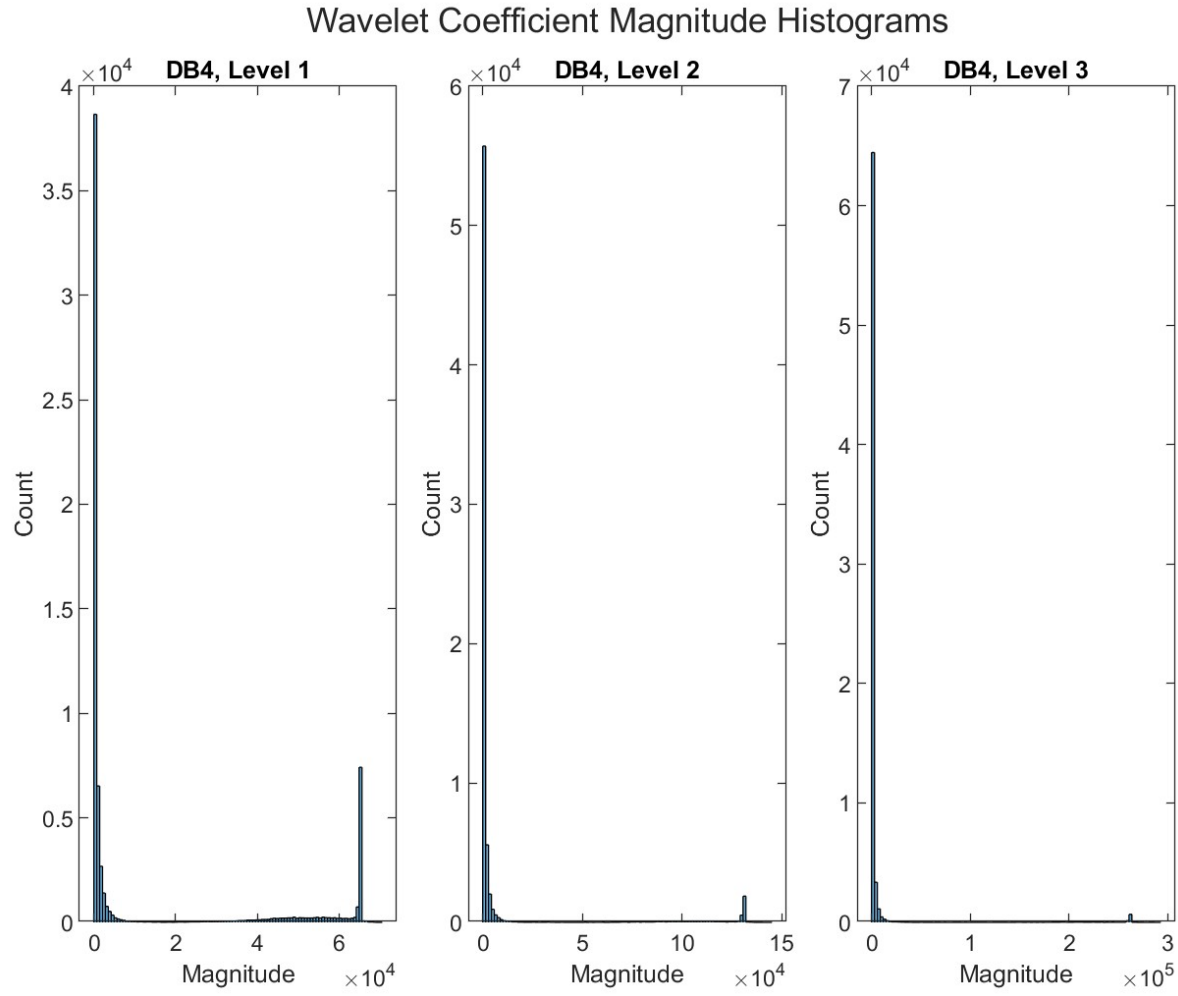


Figure 18: Wavelet coefficient magnitude histograms for the guiding image using the db4 wavelet basis at decomposition levels 1, 2, and 3.

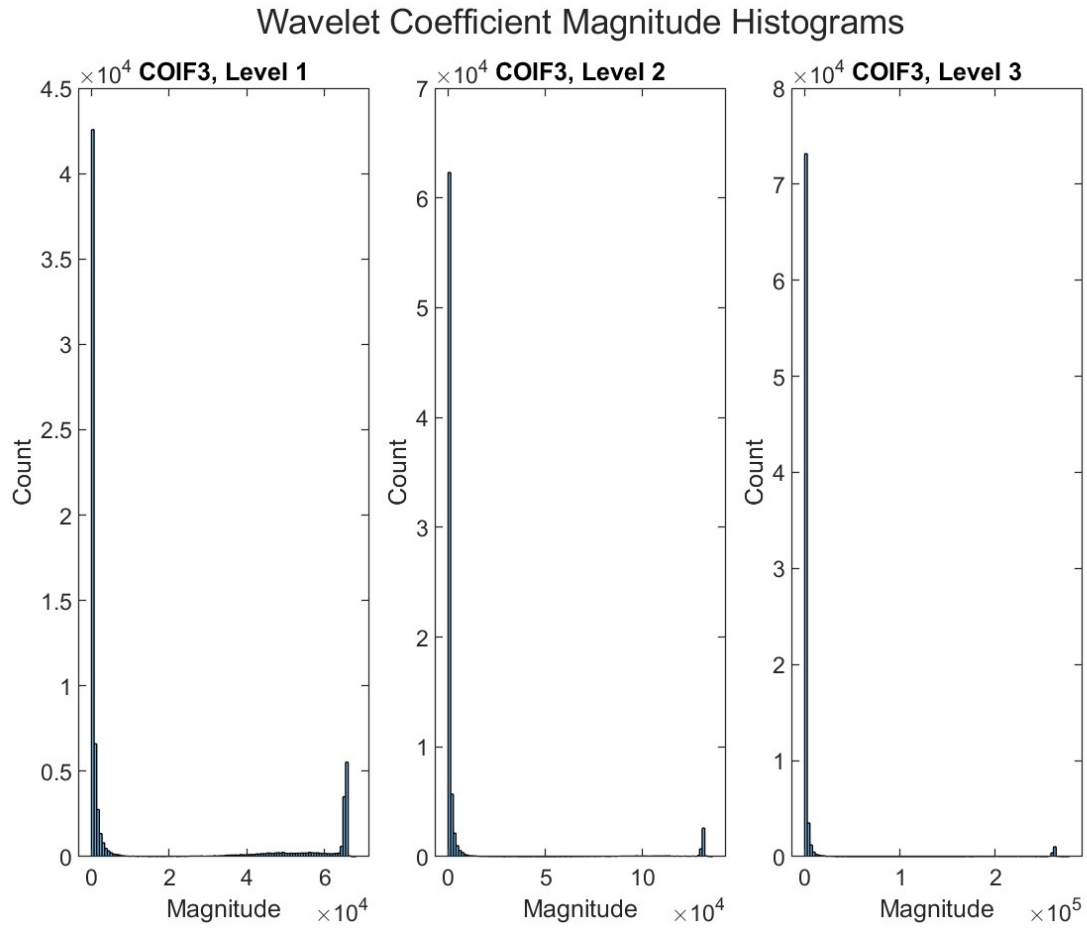


Figure 19: Wavelet coefficient magnitude histograms for the guiding image using the coif3 wavelet basis at decomposition levels 1, 2, and 3.

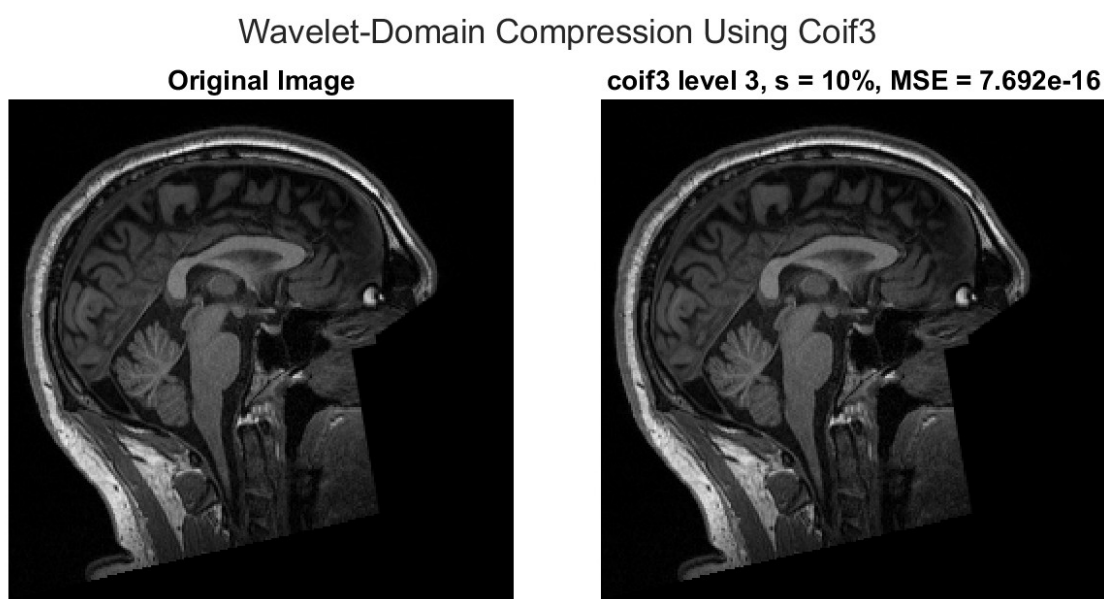


Figure 20: Comparison of the original guiding image and its wavelet-domain reconstruction using the COIF3 basis at level 3 after zeroing 10% of the smallest wavelet coefficients.

Part V: How Sparse?

Next, we evaluate how efficiently each wavelet domain reconstructs the image as increasing percentages of wavelet coefficients are removed.

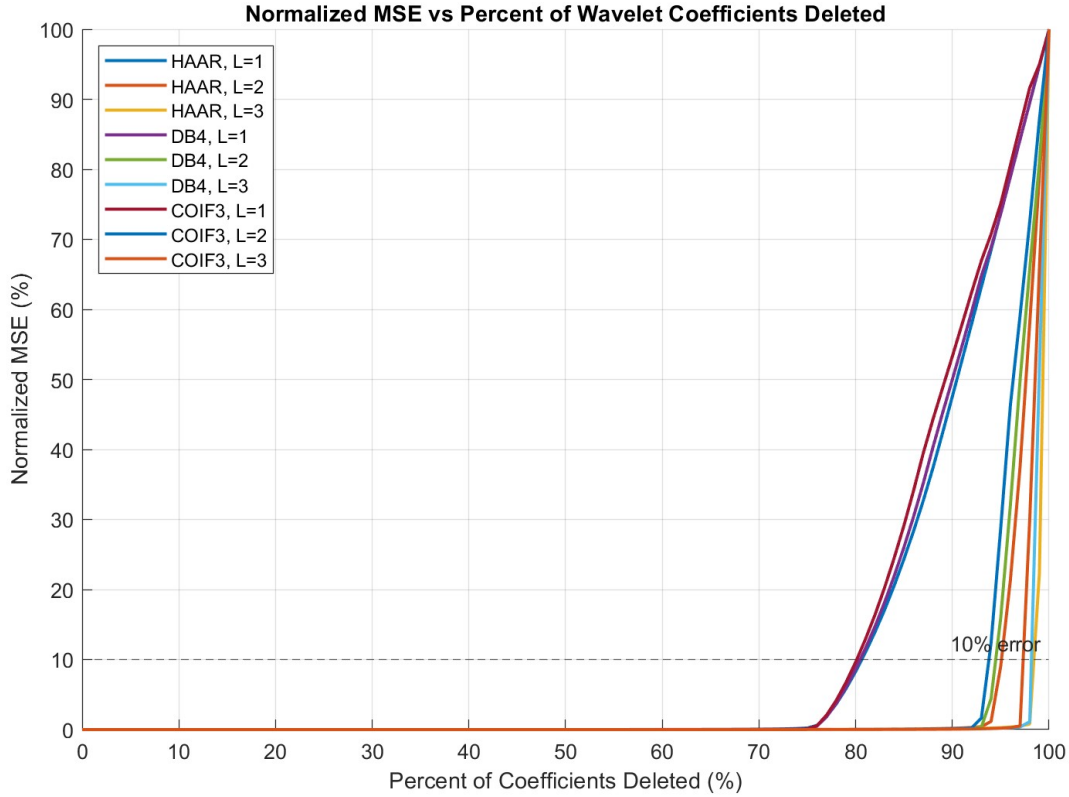


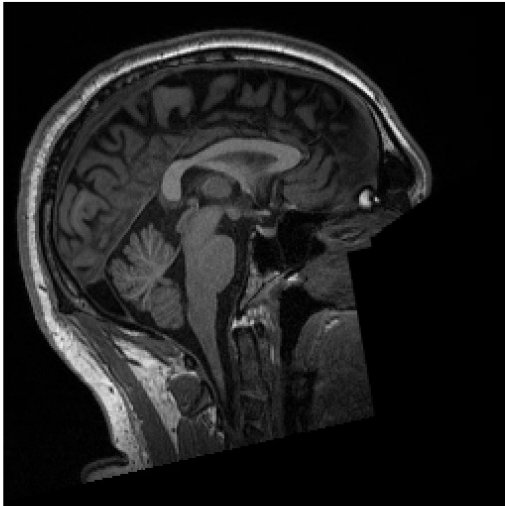
Figure 21: Percentage of Coefficients Removed (Compression) vs. Normalized MSE of the recovered guiding image across all 3 levels and 3 wavelet basis.

In Figure 21, the normalized MSE stays nearly zero for most wavelet bases and levels until a large percentage of coefficients have been removed, indicating that the guiding image is highly sparse in the tested wavelet domains. However, the error increases sharply after the deletion percentage passes a certain threshold. Lower level decompositions begin to degrade earlier, while higher level decompositions tolerate more coefficient removal before exceeding the 10% error threshold. In particular, the 3 level decompositions allow the largest fraction of coefficients to be removed while maintaining a low MSE, suggesting that these higher level wavelet decompositions provide a more efficient sparse representation.

The guiding image is reconstructed using each 3 level wavelet basis after zeroing 50% and 95% of the smallest-magnitude coefficients.

HAAR Wavelet, Level 3, 50% Coefficients Zeroed

T1, s = 50%, MSE = 0.00%



T2, s = 50%, MSE = 0.00%

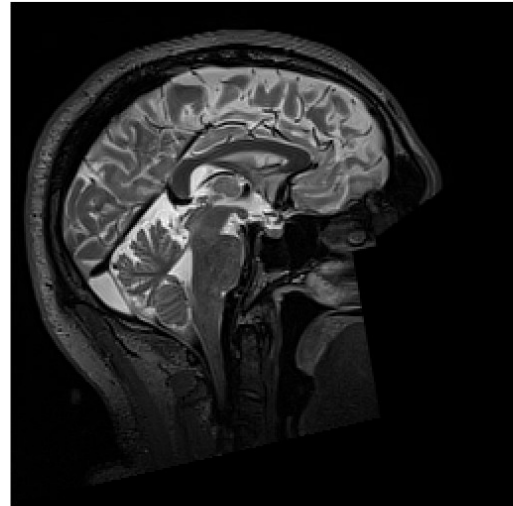
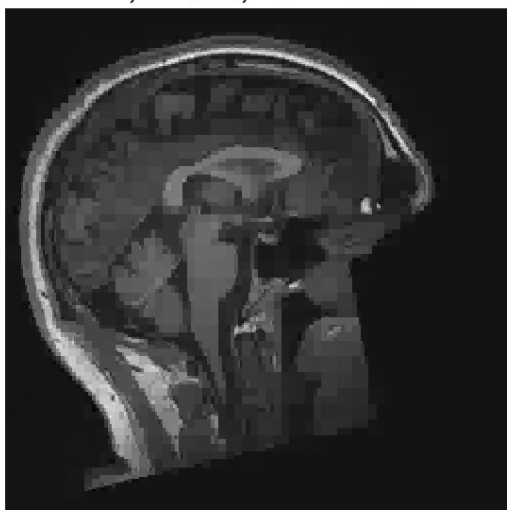


Figure 22: Reconstructed T1 and T2 guiding images at 50% compression using the 3-level Haar wavelet basis.

HAAR Wavelet, Level 3, 95% Coefficients Zeroed

T1, s = 95%, MSE = 0.27%



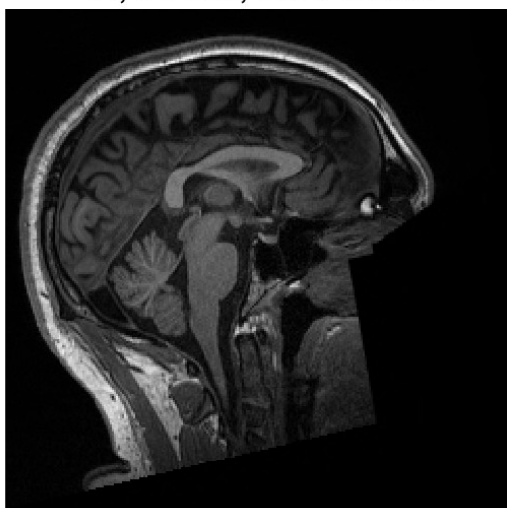
T2, s = 95%, MSE = 0.69%



Figure 23: Reconstructed T1 and T2 guiding images at 95% compression using the 3-level Haar wavelet basis.

DB4 Wavelet, Level 3, 50% Coefficients Zeroed

T1, s = 50%, MSE = 0.00%



T2, s = 50%, MSE = 0.00%

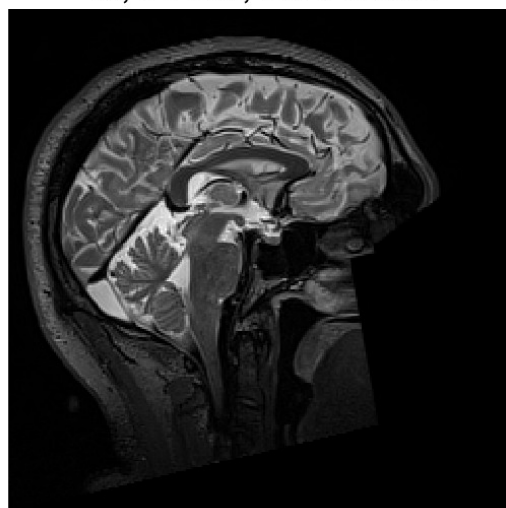


Figure 24: Reconstructed T1 and T2 guiding images at 50% compression using the 3-level db4 wavelet basis.

DB4 Wavelet, Level 3, 95% Coefficients Zeroed

T1, s = 95%, MSE = 0.19%



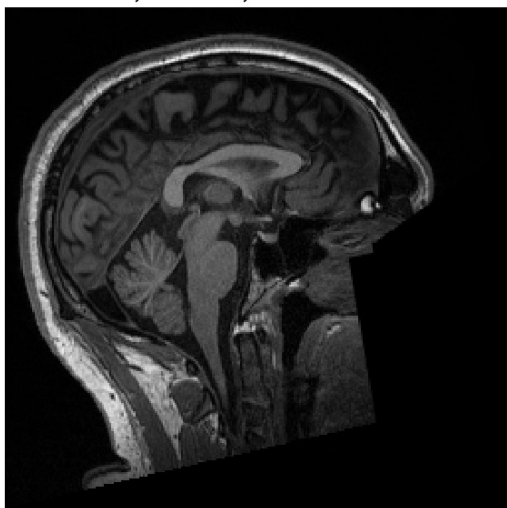
T2, s = 95%, MSE = 0.51%



Figure 25: Reconstructed T1 and T2 guiding images at 95% compression using the 3-level db4 wavelet basis.

COIF3 Wavelet, Level 3, 50% Coefficients Zeroed

T1, s = 50%, MSE = 0.00%



T2, s = 50%, MSE = 0.00%

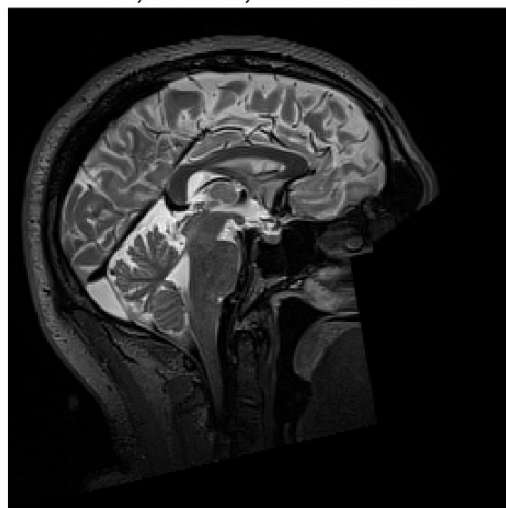
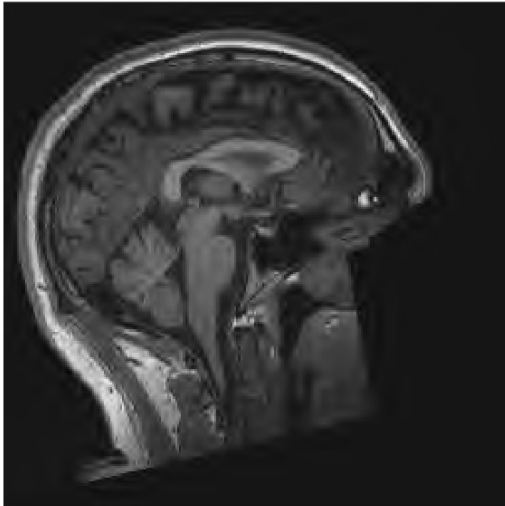


Figure 26: Reconstructed T1 and T2 guiding images at 50% compression using the 3-level coif3 wavelet basis.

COIF3 Wavelet, Level 3, 95% Coefficients Zeroed

T1, s = 95%, MSE = 0.20%



T2, s = 95%, MSE = 0.53%



Figure 27: Reconstructed T1 and T2 guiding images at 95% compression using the 3-level coif3 wavelet basis.

Qualitatively, the `db4` and `coif3` reconstructions look the best compared to the `Haar` reconstruction at both 50% and 95% loss. Quantitatively, the MSE is also lower, validating the visual inspection. It is interesting to note that in all instances, the T2 guiding image suffered much worse in terms of visual quality and MSE compared to the T1 guiding image.

Using the T1 guiding image and assuming a 10% error loss is tolerable, the maximum percentage of coefficients that can be removed is reported across all tested wavelet bases and decomposition levels. This threshold is then applied to all images to determine the maximum reconstruction error over the full dataset.

Table 1: Maximum coefficient removal for each wavelet and decomposition level under a 10% reconstruction error threshold.

Wavelet / Level	Max Coefficients Zeroed	MSE (%)
HAAR – 1	80%	8.2959
DB4 – 1	80%	8.6691
COIF3 – 1	80%	9.6498
HAAR – 2	95%	9.0499
DB4 – 2	94%	4.4095
COIF3 – 2	93%	1.6550
HAAR – 3	98%	0.8021
DB4 – 3	98%	1.1141
COIF3 – 3	97%	0.5137

This table supports the previous observation that higher level wavelet decompositions are more tolerant to compression, as they remain below the 10% error threshold even after a larger percentage of coefficients are zeroed compared to lower level decompositions.

Part VI: Compressed Sensing

A gradient descent algorithm that simulates the compressed sensing of MRI images based on sparsity in the wavelet domain is implemented. Using inspiration from the ITSA explored in class, the following parameters were selected after several trial and error runs with different λ and η values:

Table 2: Compressed Sensing Reconstruction parameters.

Parameter	Value
Sampling probabilities, p	0.1, 0.5, 0.8
Wavelet sparsity parameter, λ	0.12
Gradient descent step size, η	0.4
Number of iterations	45
Number of sensing matrices	1000
Wavelet basis	<code>coif3</code>
Decomposition level	3

Using the T1 guiding image as the ground truth, the following results for each p are:

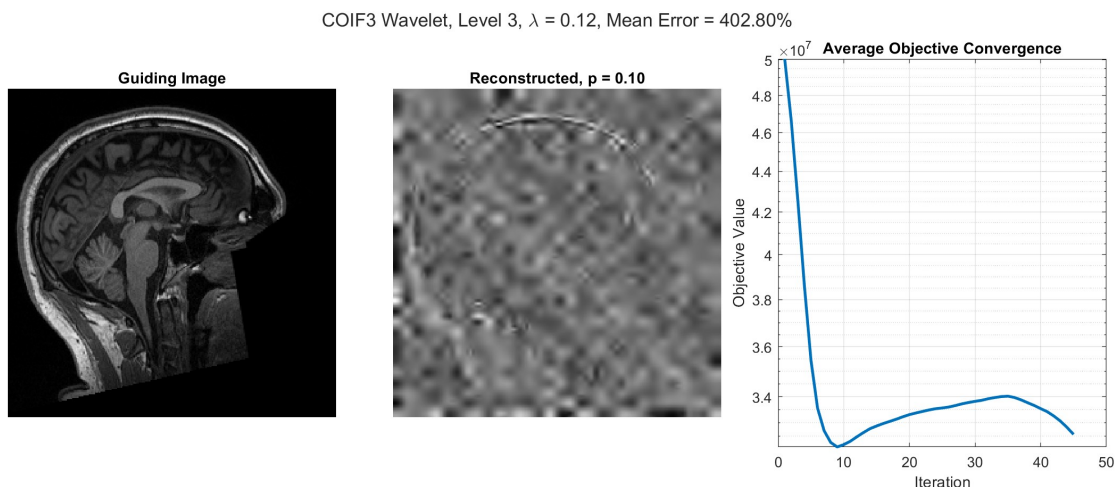


Figure 28: T1 guiding image reconstruction using a 3-level `coif3` wavelet basis with Bernoulli random undersampling at sampling probability $p = 0.10$. The left image shows the original guiding image, the center image shows the reconstruction, and the right plot shows average objective convergence.

The reconstruction was repeated for three values of the Bernoulli sampling probability p . Since p is the probability that each measurement is retained, larger values of p correspond to less aggressive undersampling. At very small values such as $p = 0.10$, the reconstruction is highly noisy and most anatomical features are lost, indicating that too few measurements are available. As p increases, the reconstructed image becomes visually clearer and the MSE decreases.

Since the goal is to preserve image quality while still reducing storage requirements, a good value of p should balance reconstruction accuracy and undersampling. Smaller values of p provide stronger compression but tend to lose important image features and increase the reconstruction error. Larger values of p preserve more image information but reduce the amount of compression achieved. Based on the MSE results, the most reasonable region to test further is $p \in [0.5, 0.8]$,

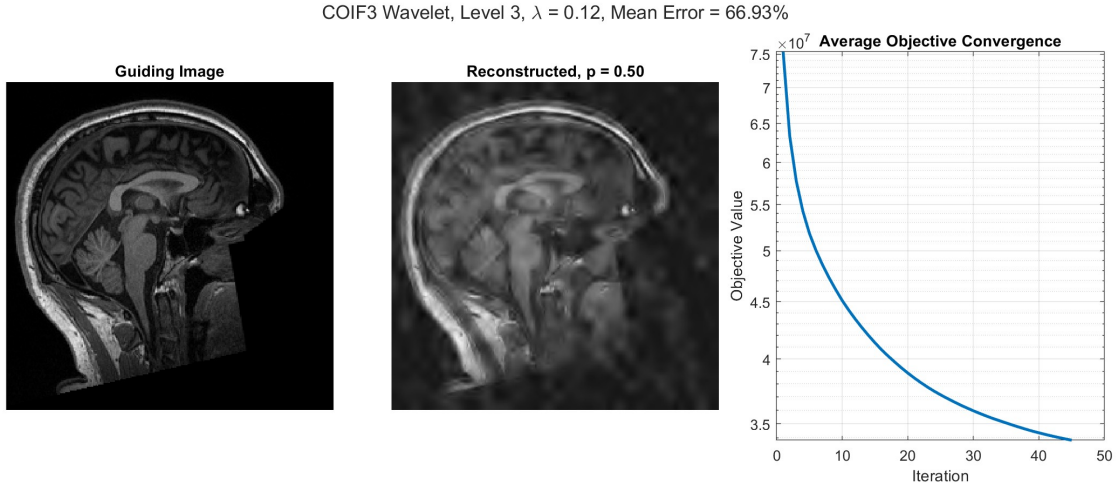


Figure 29: T1 guiding image reconstruction using a 3-level `coif3` wavelet basis with Bernoulli random undersampling at sampling probability $p = 0.50$. The left image shows the original guiding image, the center image shows the reconstruction, and the right plot shows average objective convergence.

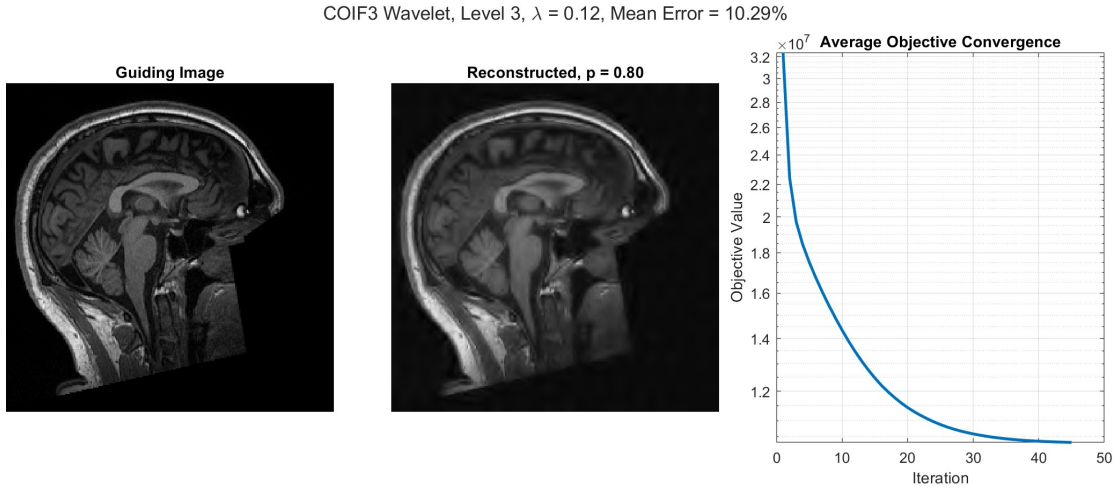


Figure 30: T1 guiding image reconstruction using a 3-level `coif3` wavelet basis with Bernoulli random undersampling at sampling probability $p = 0.80$. The left image shows the original guiding image, the center image shows the reconstruction, and the right plot shows average objective convergence.

since these values produced the lowest reconstruction errors while still retaining some level of undersampling. As with many parameter-selection problems, the final choice of p boils down to the desired tradeoff between accuracy and compression. Depending on the acceptable error threshold, desired image quality, and storage constraints, one should choose the smallest value of p that best fits all three.

Conclusion

This project demonstrated that anatomical MRI images can be represented sparsely in both Fourier and wavelet domains, enabling effective compression and reconstruction. Wavelet-based methods were introduced as an alternative to standard Fourier domain compression and produced better sparse representations with lower reconstruction error. Among the tested bases, the 3-level `coif3` wavelet basis produced the best results, preserving image quality even after significant coefficient removal. In the compressed sensing experiments, the ISTA reconstruction quality improved as the Bernoulli sampling probability p increased, demonstrating the expected tradeoff between undersampling and reconstruction accuracy. Overall, the results indicate the usefulness of wavelet sparsity for MRI compression. The choice of wavelet basis, decomposition level, and sampling probability all strongly affect reconstruction quality so it ultimately boils down to a tradeoff question depending on the engineering objective.